

Design Pattern 4: Adapter Pattern

Exercise 4: Implementing the Adapter Pattern

Scenario:

You are developing a payment processing system that needs to integrate with multiple third-party payment gateways with different interfaces. Use the Adapter Pattern to achieve this.

Steps:

Create a New Java Project:

Create a new Java project named AdapterPatternExample.

Define Target Interface:

Create an interface PaymentProcessor with methods like processPayment().

Implement Adaptee Classes:

Create classes for different payment gateways with their own methods.

Implement the Adapter Class:

Create an adapter class for each payment gateway that implements PaymentProcessor and translates the calls to the gateway-specific methods.

Test the Adapter Implementation:

Create a test class to demonstrate the use of different payment gateways through the adapter.